

Xue (Snow) Yu

Biostatistician II | Division of Nephrology, School of Medicine, Stanford University

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Publications and presentations appear under the names Xue Yu and Snow Yu.

RESEARCH INTERESTS

Statistical and computational methods for kidney and population health. Longitudinal modeling of disease progression; how model choice changes who is identified as at risk; sample representativeness and the limits of generalizability; reproducible analysis workflows; and responsible evaluation of AI in healthcare.

EDUCATION

Georgetown University

2017–2019

Master of Science, Biostatistics | Washington, DC

Master's research project: Statistical Tests for the Comparison of Several Medians. Advisor: George Luta, PhD.

Hong Kong Baptist University

2013–2017

Bachelor of Science (Honours), Financial Mathematics | Hong Kong

RESEARCH EXPERIENCE

Stanford University

Apr 2022–Present

Biostatistician II | Division of Nephrology, School of Medicine

Compared ordinary least squares, linear mixed-effects, and latent class mixed-effects models in R to identify rapid kidney function decline among 292 participants in the Kidney Progression Project, a CKDu cohort in Sri Lanka. Showed that the three approaches classify different participants as rapid decliners, and characterized where they diverge. Conducted pooled and sex-stratified analyses, interpreted censoring limitations with a faculty mentor, and wrote the first-author JSM 2024 poster.

Developed a generalizability score quantifying how closely the RADx-UP cohort matches the 2020 U.S. Census, surfacing demographic gaps that aggregate summaries conceal, together with a visualization making the gaps legible to non-statisticians. Wrote the first-author JSM 2025 poster; re-presented at the Stanford SRP Symposium with the method, rather than the cohort, as the focus.

Built reusable trial data workflows and R/SAS functions for baseline tables, standardized differences, and missing-data summaries, now used across RADx-UP analyses.

Applied multiple imputation by chained equations and generalized estimating equations accounting for facility-level clustering in a nationwide cluster randomized trial across 62 dialysis centers under the NIH RADx-UP initiative.

Designed and launched a web-based flowchart tool enabling clinical fellows without coding experience to produce publication-ready diagrams; over 200 uses between June 2025 and January 2026.

University of Mississippi Medical Center

Jun 2019–Feb 2022

Biostatistician; promoted to Biostatistician II in 2022

Department of Data Science, Bower School of Population Health

Led a first-author study of QTc duration in 293 adults and 121 children with sickle cell disease, using multivariable regression benchmarked against a published Jackson Heart Study model to test whether this population differs from the general African American population. Ran a sensitivity analysis excluding six adult diuretic users with J. F. Maher.

Analyzed amino acid profiles during sickle cell vaso-occlusive crisis using partial least squares-discriminant analysis, and contributed to genetic analyses developing a polygenic risk score for QT prolongation.

Georgetown University

Jan–Dec 2018

Research Assistant | Department of Biostatistics

Prepared data, programmed statistical analyses, and contributed to analysis plans for clinical and laboratory studies.

PEER-REVIEWED JOURNAL PUBLICATIONS

First author

Yu X, Majumdar S, Pollard JD, Jackson E, Knudson J, Wolfe D, Kato GJ, Maher JF. Clinical and Laboratory Correlates of QTc Duration in Adult and Pediatric Sickle Cell Disease. *American Journal of Medicine Open*. 2023;10:100045. doi:10.1016/j.ajmo.2023.100045

Coauthored

Elliott A, Anand S, Heller HC, Orellana E, Contreras Nieves M, Ley C, Montez Rath M, Yu X, Bevill D, Flores L, Casa D, Chicas R, Hertzberg VS, Riden H, Fathallah FA, Glaser J, Morrissey-Basler MC, Parsonnet J. Beyond Water, Rest, and Shade: Advancing Farmworker Heat Protection Through Partnership. *American Journal of Industrial Medicine*. 2026;69(9):687-695. doi:10.1002/ajim.70105

Varkila MRJ, Montez-Rath M, Yu X, Subramanian N, Owens DK, Brady B, Block GA, Parsonnet J, Chertow GM, Anand S. Dialysis Facility Closures in the US From 2018 to 2024: A Serial Cross-Sectional Study. *American Journal of Kidney Diseases*. 2026;87(6):750-760.e1. doi:10.1053/j.ajkd.2025.12.003

Contreras Nieves M, Hewapathirana S, Zabel FQ, Mahalekam D, Yu X, Weerakoon C, Bandara M, Bandara I, Maduwantha I, Hearth D, Athauda I, Rathnayake B, Erandika N, Pushpakumara N, Lee E, Baron K, Bhalla V, Artiles KL, Fire A, Schensul SL, Chandrajith R, Vlahos P, Montez-Rath M, Anand S, Nanayakkara N. Case-control study of water and agricultural exposures among women in CKDu-endemic Sri Lanka. *Journal of Water and Health*. 2026;24(2):160-175. doi:10.2166/wh.2026.175

Contreras Nieves M, Aziz A, Powers A, Banda E, Jaskulak S, Cerezo M, Ramirez I, Yu X, Moreno M, Jimenez D, Ajang I, Garcia P, Dhingra H, Krishna GG, Lunn MR, Kurella Tamura M, Montez-Rath M, Rodriguez J, Yang T, Anand S. Why Am I on Dialysis? Gaps in Rural CKD Care in Central California. *Clinical Journal of the American Society of Nephrology*. 2026;21(9):1660-1663. doi:10.2215/CJN.0000000965

Subramanian N, Anand S, Yu X, Montez-Rath ME, Jarhyan P, Rajan S, Venkateshmurthy NS, Levin A, Prabhakaran D, Craig P, Prabhakaran P, Mohan S, Jaacks L. Community-Specific Differences in Kidney Function in Rural and Urban India. *Kidney International Reports*. 2025;10(6):2035-2040. doi:10.1016/j.ekir.2025.04.006

Montez-Rath M, Varkila M, Yu X, Brillhart S, Morgan C, Leppink A, Block MS, Mehta S, Hunsader P, Fountaine A, Subramanian N, Dittrich M, Owens DK, Chertow GM, Parsonnet J, Anand S, Block GA. Acceptance of SARS-CoV-2 Surveillance Testing Among Patients Receiving Dialysis: A Cluster Randomized Trial. *JAMA Network Open*. 2024;7(9):e2434159. doi:10.1001/jamanetworkopen.2024.34159

Thule J, Yu X, Montez-Rath ME, Chertow GM. MyPlate Awareness and Engagement and Perceived and Objective Diet Quality in US Adults With Chronic Kidney Disease. *Journal of Renal Nutrition*. 2024;34(6):493-499. doi:10.1053/j.jrn.2024.04.007

Zhang M, Hillegass WB, Yu X, Majumdar S, Pollard JD, Jackson E, Knudson J, Wolfe D, Kato GJ, Maher JF, Mei H. Genetic variants and effect modifiers of QT interval prolongation in patients with sickle cell disease. *Gene*. 2024;890:147824. doi:10.1016/j.gene.2023.147824

Varkila MRJ, Montez-Rath ME, Salomon JA, Yu X, Block GA, Owens DK, Chertow GM, Parsonnet J, Anand S. Use of Wastewater Metrics to Track COVID-19 in the US. *JAMA Network Open*. 2023;6(7):e2325591. doi:10.1001/jamanetworkopen.2023.25591

Hewavitharana P, Schensul S, Lee E, Montez-Rath M, Senarathne S, Liu S, Harold K, Hewapathirana S, Erandika N, Abeysundara HTK, Yu X, Bhalla V, Fire A, Levin A, Anand S, Vlahos P, Chandrajith R, Nanayakkara N. Describing Natural History and Exploring Risk Factors for Kidney Function Decline in Persons With CKD of Uncertain Etiology in Sri Lanka. *Kidney International Reports*. 2023;8(7):1430-1438. doi:10.1016/j.ekir.2023.04.010

Anand S, Montez-Rath M, Varkila M, Yu X, Block M, Brillhart S, Leppink A, Hunsader P, Owens DK, Chertow GM, Parsonnet J, Block GA. Feasibility and Acceptability of SARS-CoV-2 Screening among Patients Receiving Hemodialysis: A Pilot Study. *Clinical Journal of the American Society of Nephrology*. 2023;18(7):930-932. doi:10.2215/CJN.0000000000000137

Zhou Y, Yu X, Nicely A, Cunningham G, Challa C, McKinley K, Nickel R, Campbell A, Darbari D, Summar M, Majumdar S. Amino acid signature during sickle cell pain crisis shows significant alterations related to nitric oxide and energy metabolism. *Molecular Genetics and Metabolism*. 2022;137(1-2):146-152. doi:10.1016/j.ymgme.2022.08.004

PREPRINT

Subramanian N, Yu X, Varkila M, Hunsader P, Morgan C, Mehta S, Dittrich M, Block G, Owens DK, Parsonnet J, Chertow GM, Anand S, Montez-Rath M. Time of Day of SARS-CoV-2 Vaccination and Antibody Response in Patients Receiving Dialysis. SSRN. Preprint posted July 23, 2025. doi:10.2139/ssrn.5337673

FIRST-AUTHOR RESEARCH POSTERS

Yu X, Varkila MRJ, Subramanian N, Parsonnet J, Chertow G, Anand S, Montez-Rath M. Demographic Representativeness of the RADx-Up Cohort Compared to the U.S. Census Data. Joint Statistical Meetings, 2025. Also presented at the Stanford SRP Poster Symposium (2025), retitled Generalizability Score to Assess Sample Representativeness: Computation and Visualization following JSM feedback.

Yu X, Charu V, Anand S, Nanayakkara N, Vlahos P, Chandrajith R, Montez-Rath M. Comparing statistical models for identifying rapid decline in chronic kidney disease. Joint Statistical Meetings, 2024. Also presented at the Stanford SRP Poster Symposium.

MENTORING AND STATISTICAL CONSULTING

Mentor, Stanford Racial Equity to Advance a Community of Health (REACH) post-baccalaureate research program. Trained a pre-medical student from Morehouse School of Medicine (M.D. Class of 2026) and a second-year biology student from UCLA/Charles Drew during summer research programs.

Mentored undergraduate researchers from the National University of Singapore and master's students from UC San Diego and Stanford Epidemiology.

Supervised student research on MyPlate awareness and diet quality in CKD, resulting in a coauthored *Journal of Renal Nutrition* article, and on AI prediction of CKDu progression, reported in the WCN 2026 abstract.

Twice-monthly biostatistical office hours for the Division since October 2022; statistical consulting for Q2BI and for clinical and statistical teams in Nanjing, China.

LEADERSHIP AND SERVICE

Quantitative Staff Network

Nov 2024–Present

Chair, Communications Subcommittee

Led newsletter development, LinkedIn outreach, survey design, member engagement, and mentorship program planning; helped organize career workshops and networking events. quantitativestaffnetwork.org

Biostats4You: reviewer of biostatistics educational resources, ensuring statistical content in papers, posters, videos, and blogs is accurate and accessible to non-statistician researchers (2023). Reviewer listing

SELECTED EDUCATIONAL PRESENTATIONS

Cosmos: A Collaborative Platform Empowering Healthcare. Kidney Clinical Research Conference, Stanford (September 25, 2025).

Missing Data and Multiple Imputation Methods. Nephrology Biostatistics group meeting, Stanford (September 27, 2022).

Variable Selection in Multivariable Models. Kidney Clinical Research Conference, Stanford (June 27, 2022).

COAUTHORED CONFERENCE ABSTRACTS AND POSTERS

Aziz A, Contreras Nieves M, Powers A, Banda E, Jaskulak S, Cerezo M, Ramirez I, Yu X, Moreno M, Jimenez D, Ajang I, Garcia P, Dhingra H, Krishna GG, Lunn MR, Kurella Tamura M, Montez-Rath M, Rodriguez J, Yang T, Anand S. Exploring Pesticide Exposure and Demographic Characteristics Among Patients Receiving Dialysis in California's Central Valley. *American Journal of Kidney Diseases*. 2026;87(5):S131-S132. National Kidney Foundation Spring Clinical Meetings 2026.

Ashktorab Y, Hewapathirana S, Yu X, Anand S, Chandrajith R, Nanyakkara N, Montez-Rath M. From algorithms to agriculture: Applying AI methods to predict kidney disease in farmworkers. *Kidney International Reports*. 2026;11:104327. World Congress of Nephrology 2026, abstract WCN26-3861. doi:10.1016/j.ekir.2026.104327

Contreras Nieves M, Hewapathirana S, Zabel FQ, Yu X, Lee E, Baron K, Vlahos P, Chandrajith R, Montez Rath M, Anand S, Nanayakkara N. Case-control study characterizing drinking water and agricultural practices among women in kidney disease endemic and non-endemic regions in Sri Lanka. *Kidney International Reports*. 2026;11:106006. World Congress of Nephrology 2026, poster WCN26-3918. doi:10.1016/j.ekir.2026.106006

Contreras Nieves M, Hewapathirana S, Zabel FQ, Yu X, Baron KD, Vlahos P, Montez-Rath ME, Anand S, Nanayakkara N. Zooming In on the Outliers: A Case-Control Study Characterizing Drinking Water and Agricultural Practices Among Women in Kidney Disease Endemic and Nonendemic Regions in Sri Lanka. *ASN Kidney Week* 2025. Abstract TH-PO1029. doi:10.1681/ASN.2025j31m5hfw

Subramanian N, Yu X, Varkila M, Block GA, Parsonnet J, Chertow GM, Anand S, Montez-Rath ME. Vaccination Strategies in Patients Receiving Dialysis: Should We Watch the Clock? *ASN Kidney Week* 2024. Abstract TH-PO1149. doi:10.1681/ASN.2024swtr2jc

Varkila M, Montez-Rath ME, Yu X, Subramanian N, Chertow GM, Parsonnet J, Anand S. Trends in Dialysis Facility Access in the United States from 2018–2023 and Association with Area Socioeconomic Disadvantage. *ASN Kidney Week* 2024. Abstract TH-OR05. doi:10.1681/ASN.2024ak9m2bac

Contreras Nieves M, Powers AJ, Jaskulak-Gonzalez SA, Banda E, Yu X, Kurella Tamura M, Anand S. Why Am I on Dialysis? Exploring the Gap between Patient Perspectives and Clinical Diagnosis of Kidney Disease in California's Central Valley. *ASN Kidney Week* 2024. Abstract FR-PO1034. doi:10.1681/ASN.2024axxdzryb

Subramanian N, Prabhakaran P, Venkateshmurthy NS, Jaacks LM, Mohan S, Jarhyan P, Craig P, Yu X, Prabhakaran D, Montez-Rath M, Anand S. Kidney function within Indian populations: variation by geography and population-density. World Congress of Nephrology 2024, abstract WCN24-2024. Kidney International Reports. 2024;9(4):S650. doi:10.1016/j.ekir.2024.02.1366

Montez-Rath ME, Varkila M, Yu X, Parsonnet J, Chertow G, Block GA, Anand S. SARS-CoV-2 Testing During Routine Hemodialysis Care: A Nationwide Pragmatic Clinical Trial. ASN Kidney Week 2023. Abstract TH-PO1132. Conference record

Hewapathirana S, Lee E, Weerakoon C, Erandika N, Herath D, Bandara I, Pushpakumara N, Bandara M, Harold K, Yu X, Montez Rath M, Vlahos P, Chandrajith R, Anand S, Nannayakkara N. Approach to early diagnosis of CKDu by assessing an at risk community in Sri Lanka. World Congress of Nephrology 2023, abstract WCN23-0956. Kidney International Reports. 2023;8(3):S125. doi:10.1016/j.ekir.2023.02.283

Contreras Nieves M, Garcia P, Jimenez-Briseño D, Ajang I, Yu X, Montez-Rath M, Anand S. End-Stage Kidney Disease of Unknown Etiology: Studying a Hotspot in California's Central Valley. 43rd Annual Dialysis Conference, February 2023. Conference record

Noor R, Akam-Venkata J, Joseph M, Knudson J, Lirette ST, Yu S, Majumdar S, Shakti D. Abnormal left ventricular strain in children with sickle cell disease. American Society of Echocardiography Scientific Sessions, June 10–13, 2022; Seattle, WA.

Tio M, Yu X, Hall ME, Shafi T. A comparison of trends in chronic kidney disease in Mississippi using the Chronic Kidney Disease Epidemiology Collaboration 2009 vs. 2021 equations. Southern Regional Meeting, February 10–12, 2022; New Orleans, LA. Journal of Investigative Medicine. 2022;70(2):740-741. Conference record

TECHNICAL SKILLS

Programming and reproducible research: R, SAS, Stata, Python, SQL, R Markdown, R Shiny, Git, LaTeX. Methods: regression, mixed-effects models, survival analysis, GEE, predictive modeling, missing-data imputation, latent class modeling. Research systems: Epic Cosmos, REDCap, Qualtrics, Tableau.

PROFESSIONAL TRAINING AND CREDENTIALS

Epic Cosmos: Super User Badge (August 2026); Researcher Badge (May 2026); Data Model Accreditation (March 2025).

Stanford coursework (audited / non-credit): BIOMEDIN/BMDS 223, Deploying and Evaluating Fair AI in Healthcare (Spring 2026); EPI 262, Intermediate Biostatistics: Regression, Prediction, and Survival Analysis (Spring 2024).

Professional development: Spectrum/Stanford Essentials of Clinical Research (January–March 2025); LLMs for Statisticians and Data Scientists; Fundamentals of AI/ML in Precision Medicine; Team Science Principles for Data Scientists, Statistical Horizons (September 2023).

SAS Certified Base Programmer for SAS 9 (April 2019).

HONORS AND AWARDS

National Endeavor Scholarship, Government of China (2016). Annual Merit Award for Student Ambassador, Hong Kong Baptist University (2015).

SELECTED GRADUATE COURSEWORK

Machine Learning for Bioinformatics; Data Science; Statistical Inference; Survival Analysis; Linear Models and Multivariate Analysis; Categorical Data Analysis; Experimental Design and Clinical Trials.